

MARK SCHEME for the May/June 2014 series

9709 MATHEMATICS

9709/11

Paper 1, maximum raw mark 75

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the May/June 2014 series for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level components and some Ordinary Level components.

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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more “method” steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol ∇ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously “correct” answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no “follow through” from a previous error is allowed)
CWO	Correct Working Only - often written by a “fortuitous” answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR–1	A penalty of MR–1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become “follow through” marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR–2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA–1	This is deducted from A or B marks in the case of premature approximation. The PA–1 penalty is usually discussed at the meeting.

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1	$a = 1, \quad b = 2$	B1B1 [2]	Or $1 + 2 \sin x$
2	(i) $(2x-3)^2 - 9$ (ii) $2x-3 > 4 \quad 2x-3 < -4$ $x > 3\frac{1}{2}$ (or) $x < -\frac{1}{2}$ cao Allow $-\frac{1}{2} > x > 3\frac{1}{2}$ OR $4x^2 - 12x - 7 \rightarrow (2x-7)(2x+1)$ $x > 3\frac{1}{2}$ (or) $x < -\frac{1}{2}$ cao Allow $-\frac{1}{2} > x > 3\frac{1}{2}$	B1B1 [2] M1 A1 M1 A1 [2]	For -3 and -9 At least one of these statements Allow 'and' $3\frac{1}{2}, -\frac{1}{2}$ soi scores first M1 Attempt to solve 3-term quadratic Allow 'and' $3\frac{1}{2}, -\frac{1}{2}$ soi scores first M1
3	$[{}^8C_6 \text{ or } 28] \times [16 \text{ or } 4^2] (x^6) \times \left[\frac{1}{(64 \text{ or } 2^6)(x^6)} \right]$ 7	B1B1B1 B1 [4]	Seen in expansion ok. Allow 8C_2 Identified as answer
4	$\frac{dy}{dx} = [-2 \times 4(3x+1)^{-3}] \times [3]$ When $x = -1, \frac{dy}{dx} = 3$ When $x = -1, y = 1$ soi $y - 1 = 3(x + 1) \quad (\rightarrow y = 3x + 4)$	B1B1 B1 B1 B1 ✓ [5]	$[-2 \times 4u^{-3}] \times [3]$ is B0B1 unless resolved Ft on <i>their</i> '3' only (not $-\frac{1}{3}$). Dep on diffn
5	(i) $200/2(2a + 199d) = 4 \times 100/2(2a + 99d)$ $d = 2a$ cao (ii) $a + 99d = a + 99 \times 2a$ $199a$ cao	M1A1 A1 [3] M1 A1 [2]	Correct formula used (once) M1, correct eqn A1 Sub. <i>their</i> part(i) into correct formula
6	(i) area $\Delta = \frac{1}{2} \times 4 \times 4 \tan \alpha$ oe soi Area sector $= \frac{1}{2} \times 2^2 \alpha$ oe soi Shaded area $= 8 \tan \alpha - 2\alpha$ cao (ii) $DC = \frac{4}{\cos \alpha} - 2$ oe soi Arc $DE = 2\alpha$ soi anywhere provided clear Perimeter $= \frac{4}{\cos \alpha} + 4 \tan \alpha + 2\alpha$ cao	B1 B1 B1 [3] B1 B1 B1 [3]	$4 \tan \alpha = \sqrt{16 / \cos^2 \alpha - 16}$. (Can also score in answer) Accept θ throughout Little/no working – accept terms in answer $\frac{4}{\cos \alpha} = \sqrt{16 + 16 \tan^2 \alpha}$. Can score in answer Little/no working – accept terms in answer

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7 $(a-3)^2 + (2-b)^2 = 125$ oe $\frac{2-b}{a-3} = 2$ oe $(a-3)^2 + (2a-6)^2 = 125$ (sub for a or b) $(5)(a+2)(a-8) (=0)$ Attempt factorise/solve $a = -2$ or 8, $b = 12$ or -8	B1 B1 M1 M1 A1A1 [6]	Or $1/4(2-b)^2 + (2-b)^2 = 125$ Or $(5)(b-12)(b+8) (=0)$ Answers (no working) after 2 correct eqns score SCB1B1 for each correct pair (a, b)
8 (i) $OA \cdot OB = -3p^2 - 4 + p^4$ soi $(p^2 + 1)(p^2 - 4) = 0$ oe e.g. with substitution $p = \pm 2$ and no other real solutions (ii) $\overrightarrow{BA} = \begin{pmatrix} 9 \\ 4 \\ 9 \end{pmatrix} - \begin{pmatrix} -3 \\ -1 \\ 9 \end{pmatrix} = \begin{pmatrix} 12 \\ 5 \\ 0 \end{pmatrix}$ $\overrightarrow{BA} = \sqrt{12^2 + 5^2} = 13$ and division by <i>their</i> 13 Unit vector $= \frac{1}{13} \begin{pmatrix} 12 \\ 5 \\ 0 \end{pmatrix}$ cao	M1 M1 A1 [3] M1 M1 A1 [3]	Put $= 0$ (soi) and attempt to solve Reversed subtraction can score M1M1A0
9 (i) $LHS \equiv \frac{\sin^2 \theta - (1 - \cos \theta)}{(1 - \cos \theta) \sin \theta}$ cao $\equiv \frac{1 - \cos^2 \theta - 1 + \cos \theta}{(1 - \cos \theta) \sin \theta}$ $\equiv \frac{\cos \theta (1 - \cos \theta)}{(1 - \cos \theta) \sin \theta}$ $\equiv \frac{1}{\tan \theta}$ (ii) $\tan \theta = (\pm) \frac{1}{2}$ $26.6^\circ, 153.4^\circ$	B1 M1 M1 A1 [4] M1 A1A1 $\sqrt{h}$ [3]	Put over common denominator Use $s^2 = 1 - c^2$ oe Correct factorisation from line 2 AG Ft for $180 - 1^{st}$ answer

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10 (i) $-5 \leq f(x) \leq 4$ For $f(x)$ allow x or y; allow $<$, $[-5, 4]$, $(-5, 4)$ (ii) $f^{-1}(x)$ approximately correct (independent of f) Closed region between $(1, 1)$ and $(4, 4)$; line reaches x-axis (iii) LINE: $f^{-1}(x) = \frac{1}{3}(x+2)$ for $-5 \leq x \leq 1$ CURVE: $5 - y = \frac{4}{x}$ OR $x = 5 - \frac{4}{y}$ $f^{-1}(x) = 5 - \frac{4}{x}$ oe for $1 < x \leq 4$	B1 [1] B1 DB1 [2] B1 B1B1 M1 A1 B1 [6]	Allow less explicit answers (eg $-5 \rightarrow 4$) Ignore line $y = x$ Allow $y = \dots$ but must be a function of x cao but allow $<$ cao cao but allow $<$ or $<$
11 (i) $x^2 + 4x + c - 8 (= 0)$ $16 - 4(c - 8) = 0$ $c = 12$ OR $-2 - 2x = 2 \rightarrow x = (-2)$ $-4 + c = 8 + 4 - 4$ $c = 12$ (ii) $x^2 + 4x + 3 \rightarrow (x + 1)(x + 3) (= 0) \rightarrow$ $x = -1$ or -3 $\int (8 - 2x - x^2) - [\int (2x + 11) \text{ or area of trapezium}]$ $\left[8x - x^2 - \frac{x^3}{3} \right] - \left[x^2 + 11x \right] \text{ or } \left[8x - x^2 - \frac{x^3}{3} \right] - \frac{1}{2}(5+9) \times 2$ Apply <i>their</i> limits to at least integral for curve $1\frac{1}{3}$ oe	M1 M1 A1 M1 M1 A1 [3] B1 M1M1 A1B1 M1 A1 [7]	Attempt to simplify to 3-term quadratic Apply $b^2 - 4ac = 0$. '$= 0$' soi Equate derivs of curve and line. Expect $x = -2$ Sub <i>their</i> $x = -2$ into line and curve, and equate Attempt to integrate. At some stage subtract A1 for curve, B1 for line OR $\left[-3x - 2x^2 - \frac{x^3}{3} \right] \text{ A2, 1, 0}$ For M marks allow reversed limits and/or subtraction of areas but then final A0

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12 (i) $y = \frac{2}{3}x^{\frac{3}{2}} - 2x^{\frac{1}{2}} + (c)$ oe $\frac{2}{3} = \frac{16}{3} - 4 + c$ $c = -\frac{2}{3}$ (ii) $\frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}x^{-\frac{3}{2}}$ oe (iii) $x^{\frac{1}{2}} - x^{-\frac{1}{2}} = 0 \rightarrow \frac{x-1}{\sqrt{x}} = 0$ $x = 1$ When $x = 1$, $y = \frac{2}{3} - 2 - \frac{2}{3} = -2$ When $x = 1$, $\frac{d^2y}{dx^2}(=1) > 0$ Hence minimum	B1B1 M1 A1 [4] B1B1 [2] M1 A1 M1A1 B1 [5]	Attempt to integrate Sub $\left(4, \frac{2}{3}\right)$. Dependent on c present Equate to zero and attempt to solve Sub. <i>their</i> '1' into <i>their</i> 'y' Everything correct on final line. Also dep on correct (ii). Accept other valid methods
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